

Figure 9.10-18 Example of an operator loosening the hexagonal retaining bolts.

3. Check that the clamping plate is loose



Figure 9.10-19 Example of an operator checking that the clamping plate is loose.

4. Either remove the drum edge shield or support it while removing the bolts from the retaining nuts



Figure 9.10-20 Example of an operator removing the support strip.

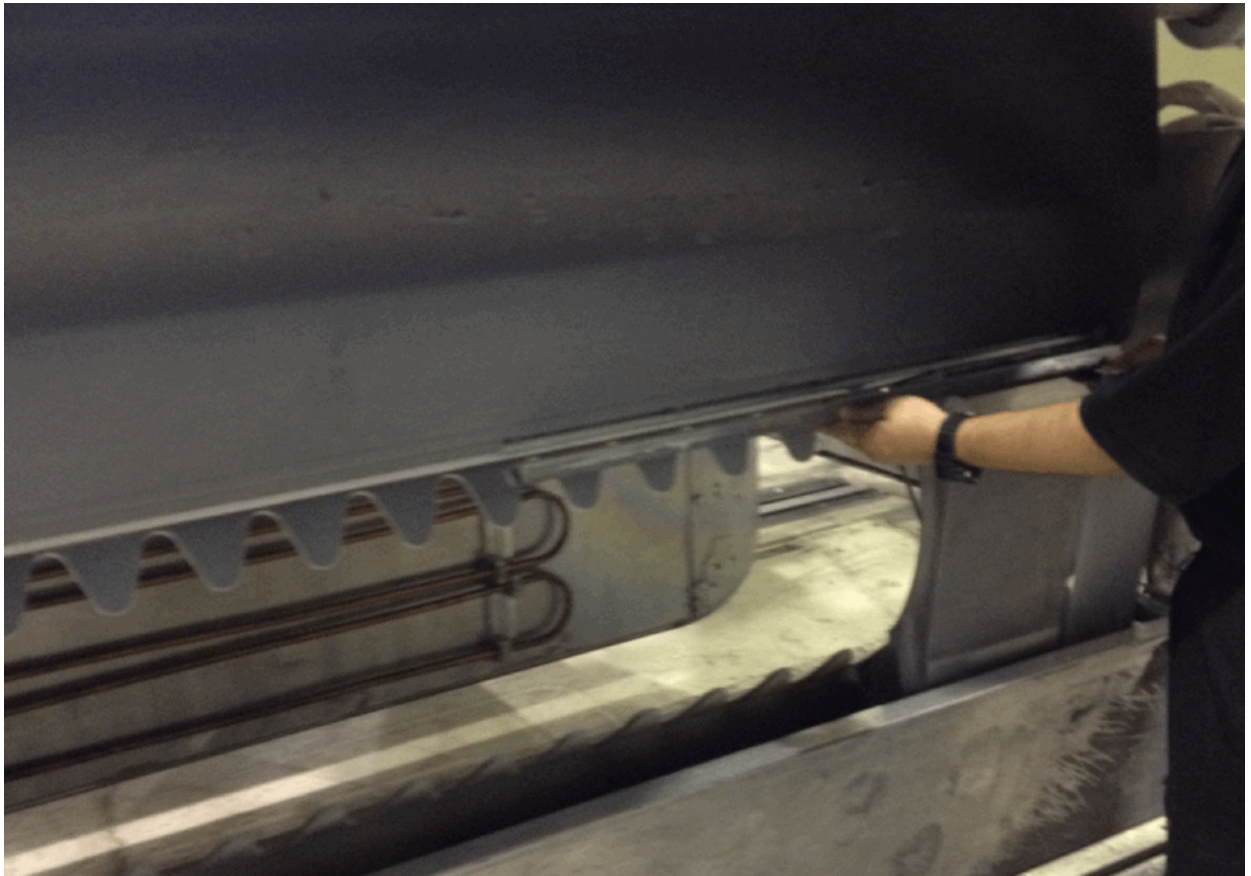


Figure 9.10-21 Example of an operator removing the support strip.



Figure 9.10-22 Example of a coating shield with the support strip removed.

|                                                                                   |                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|
|  | <b>Note !</b>                                                       |
|                                                                                   | <b>Do not use any excessive force or damage any of the surfaces</b> |

### Replacement of the Support Strip

1. Ensure that all contact surfaces are clean of any aluminium debris.
2. Visually check that the retaining channel nuts (BML part # 421791) are in position and of good condition. If not replace them. A suggestion is to mark the position using a marker pen before replacing them.
3. Install the strips and install the retaining bolts.
4. Ensure that the bolts are tightened sufficiently to the designed torque rating.

|                                                                                   |                                                                                                                     |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                                       |
|                                                                                   | <b>For ordering spare parts, check the assembly drawings supplied and the BOM to ensure correct codes are used.</b> |

## 9.11 Coating Shutter Operation

A water-cooled aluminium shutter (moving shield) is located under the coating shield, to stop evaporated aluminium reaching the substrate when there is no movement of it through the machine.

This shutter is moved by "rod less" cylinders located each side of the mechanism and supported on tracks

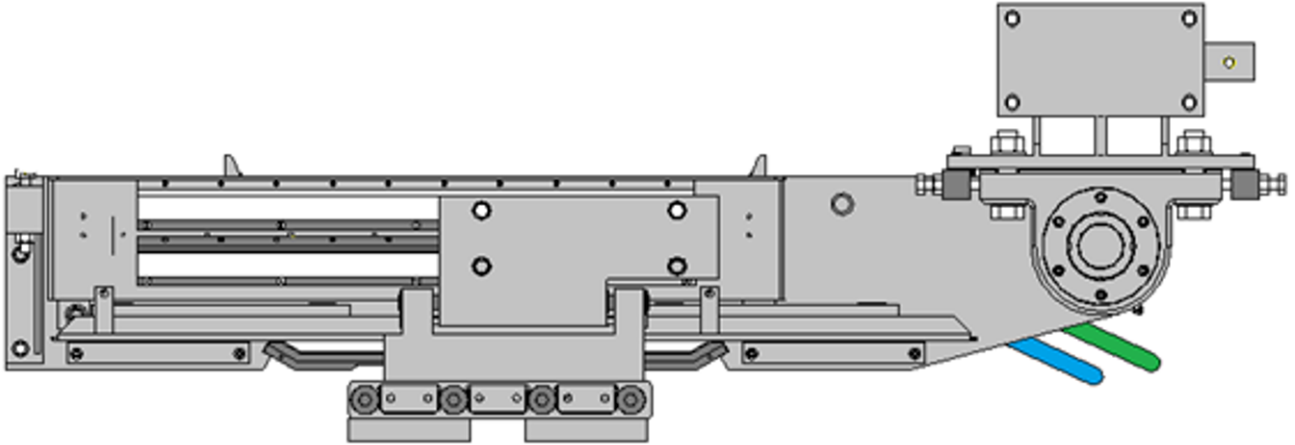


Figure 9.11-1 View of a shutter assembly layout

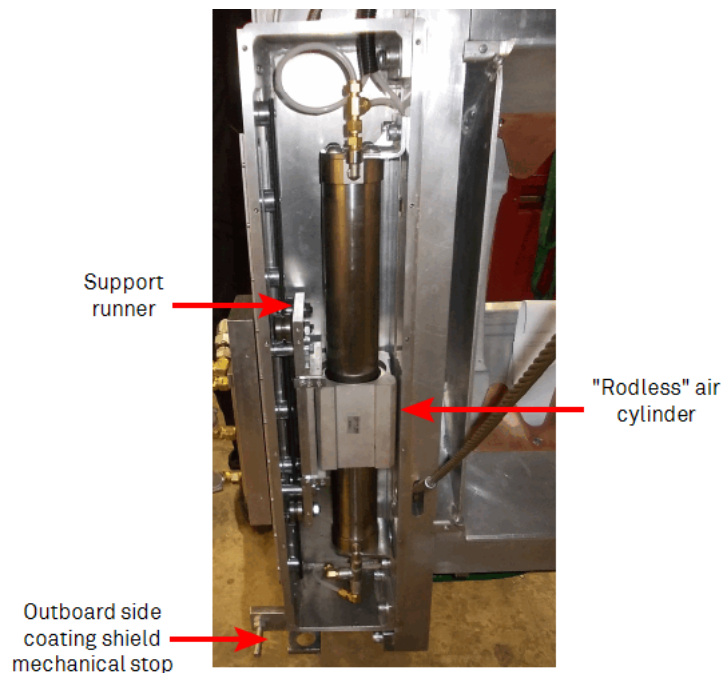


Figure 9.11-2 The shutter assembly with the rear cover removed

The machine is designed to have the shutter closed at all times unless the coating of the substrate is taking place. In the cases of a web break, the shutter will close automatically.

The shutter is interlocked to only open when a web-running signal and the operator touches the open push-button on the PROCESS screen.



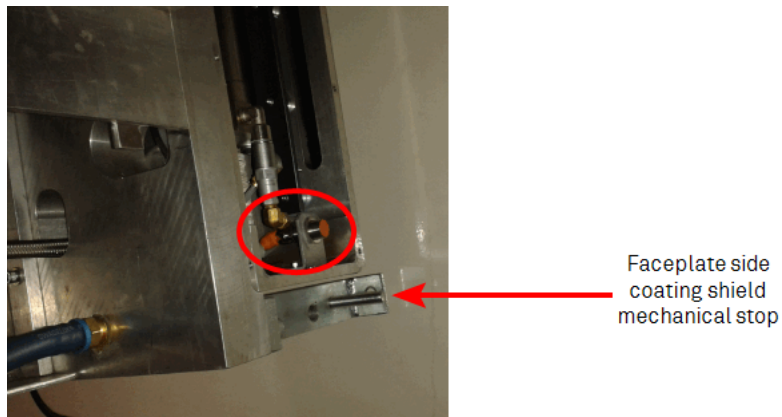


Figure 9.11-3 The shutter open limit sensor at faceplate side of shield assembly

The winding cart cannot be driven into the chamber unless the shutter is closed.

## Rodless Cylinder Pneumatic Pipe work

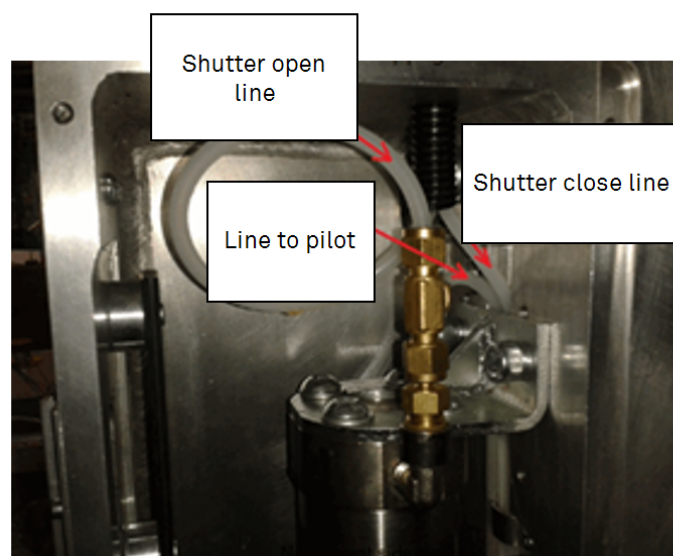


Figure 9.11-4 The upper pivot side of assembly showing pneumatic lines and components.

|                                                                                     |                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Note !</b>                                                                                                                                                                        |
|                                                                                     | <p>Use 6mm 'nylon' flexible tubing, not polyurethane, for the correct rating to be used within the vacuum system.</p> <p>All connections are to be through compression fittings.</p> |

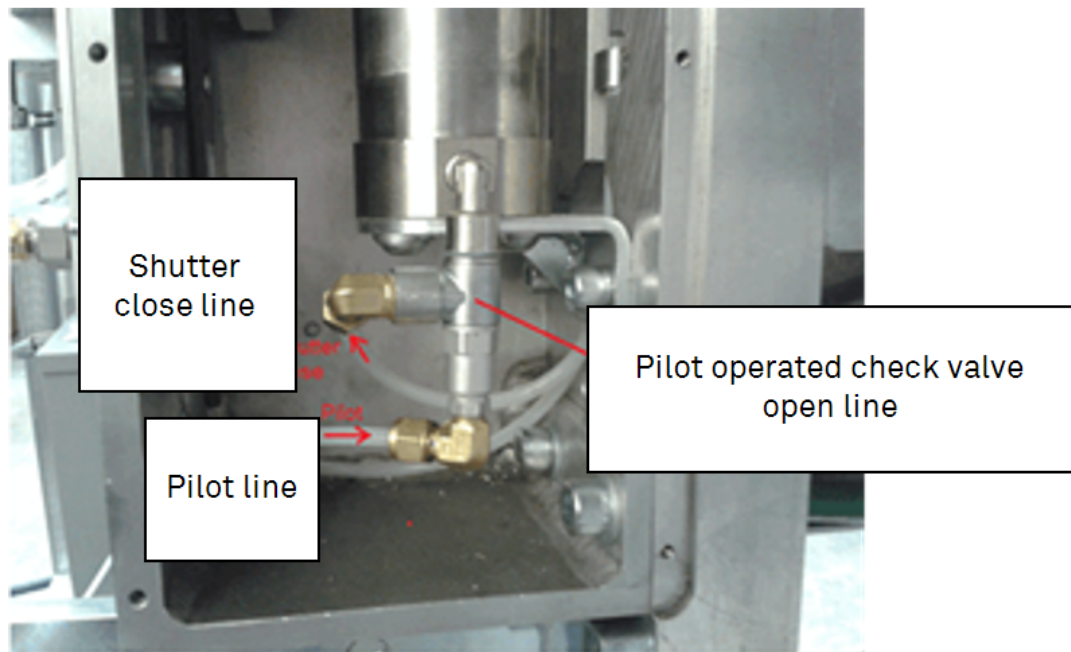


Figure 9.11-5 The lower non pivot side of assembly showing pneumatic lines and components.

This is the same arrangement for both cylinders used.

Airlines from face-plate leadthrough are routed through the partition bulk-head bracket



Figure 9.11-6 The distribution lines in the partition bulkhead

As part of the design, the air-lines are installed into 6mm tee fittings, at the centre-line of the coating shield. Ensure that the flexible tubing from these compression tee fittings to the cylinders are cut to the same length.

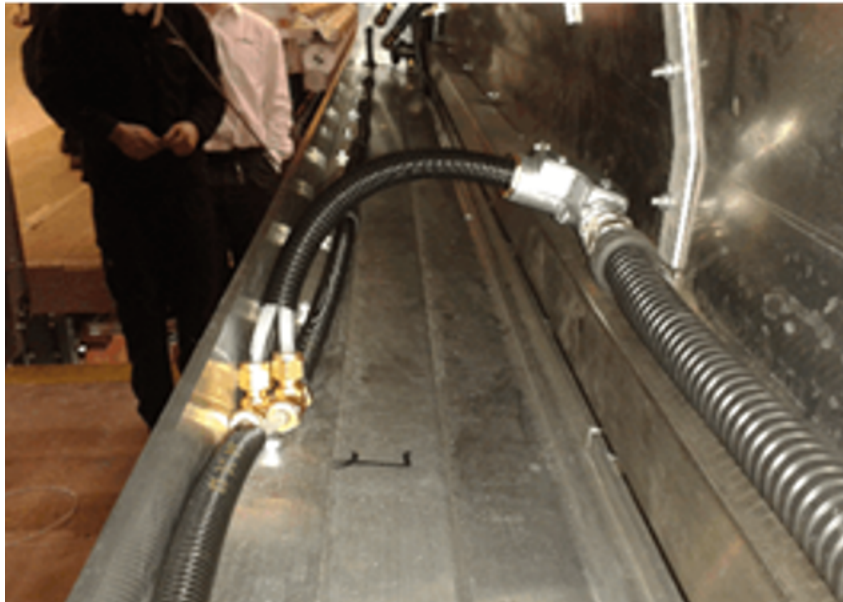


Figure 9.11-7 The airline distribution lines at the rear of the coating shield.

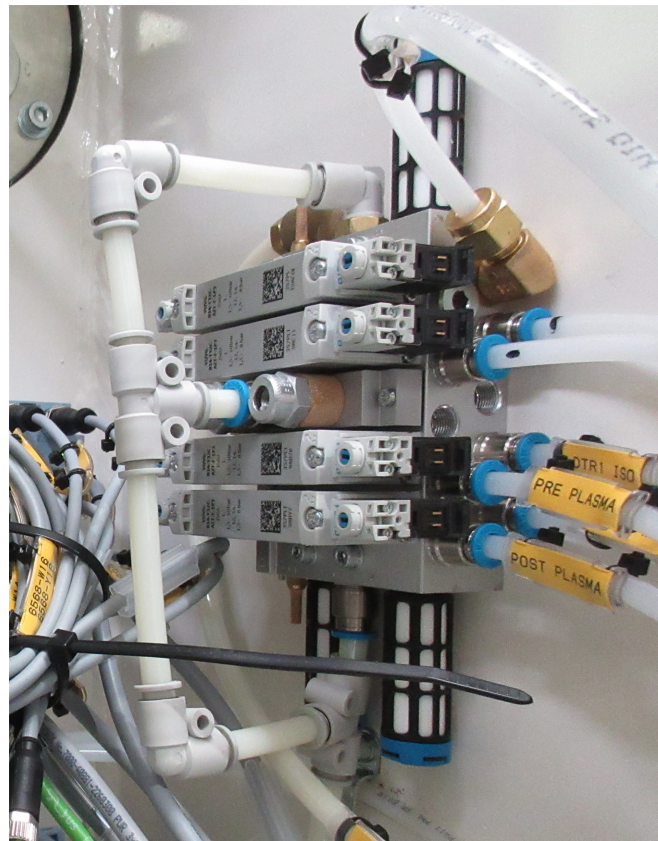


Figure 9.11-8 The external control valve arrangement on the faceplate and the lines that go through the faceplate.

## Shutter Position on the Coating Shield

Set the shutter position to 17.5 mm from the end of the shield arms, as shown below. This will allow approximately a 10mm gap between the shutter edge and the source viewing flap, when it is closed.

With air applied and the shutter moved to the open position, the final adjustment is made as shown below to achieve this dimension.

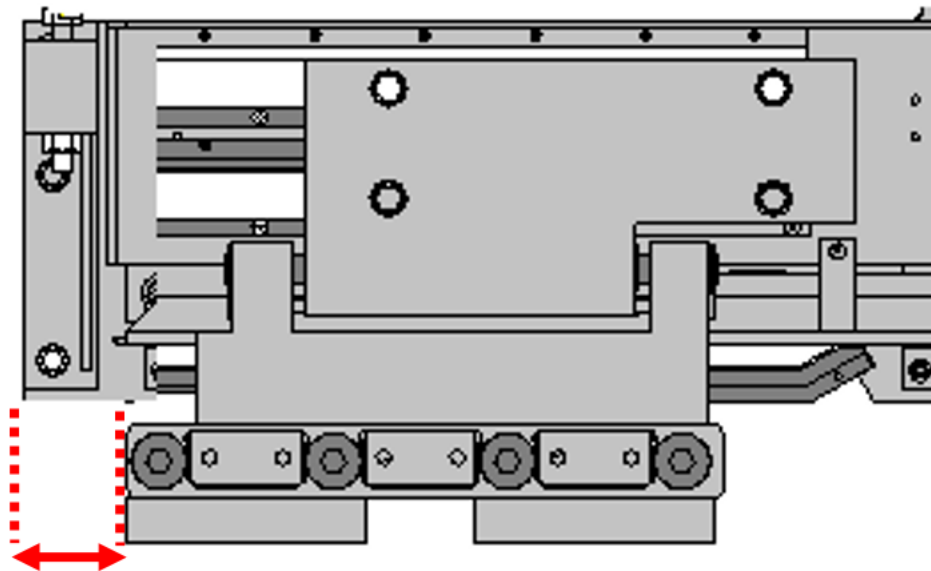


Figure 9.11-9 The required positional alignment.

The inboard (Faceplate) side is adjusted using 2 opposing jacking bolts (cap-heads) shown below



Figure 9.11-10 The shutter position adjustment bolts on the faceplate side

The outboard side should be used to remove any skew in the assembly, as there is more adjustment on this side

The outboard end is adjusted by loosening the 3 cap-heads and adjusting the 2 opposing jacking bolts.

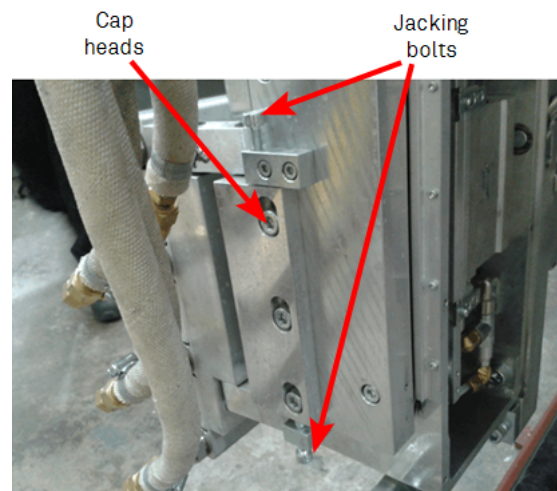


Figure 9.11-11 The shutter position adjustment bolts on the outboard side

## Shutter Vee-Rail and Cylinder Removal Instructions

This basic procedure is written with the understanding that the persons carrying out this task are competent in the skills required.



1. Lower the coating shield to its vertical position, and move the shutter to the open position (i.e. at its bottom vertical position).

|                                                                                   |                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Warning !</b>                                                                                                                                                                                                         |
|                                                                                   | <p><b>Ensure the machine is in a safe condition and all air has been removed from the system.</b></p> <p><b>Lockout the energy sources, to prevent the possibility of these sources of energy being switched on.</b></p> |

It is possible to change the cylinders and guide rails without removing the shutter, however it is advisable that the weight of the shutter is supported during this work.

2. Remove the rear access covers to gain access to the internal shutter guide assembly.
3. Remove the two button head slotted screws holding the guide plate in place, as shown.

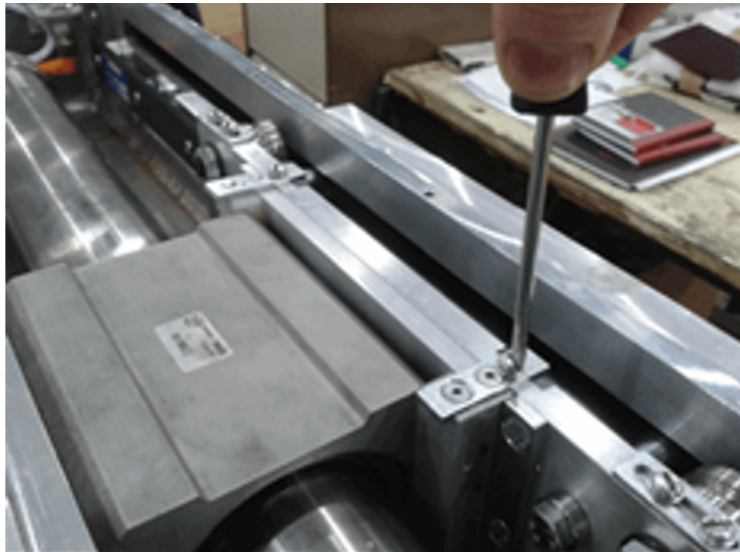


Figure 9.11-12 Example of an technician removing the button-heads screws.

4. Remove the guide plate from between the cylinder and the slide. There may be shims located between the cylinder and the guide plate. Identify them for reinstalling later

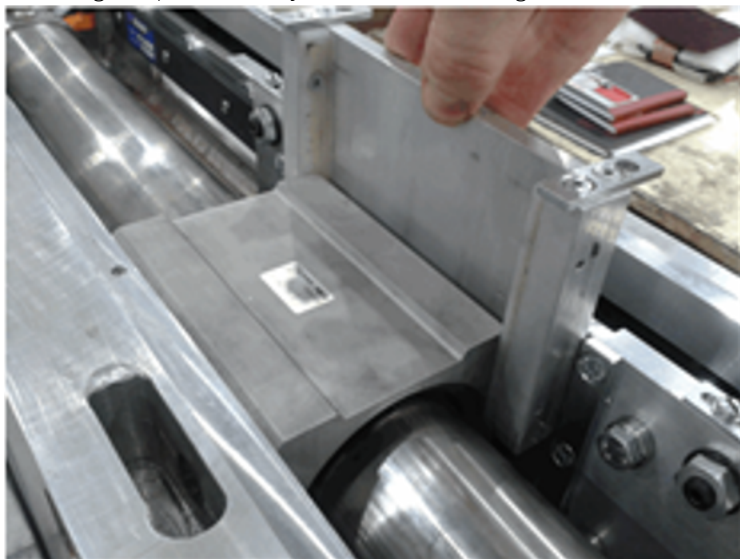


Figure 9.11-13 Example of an technician removing the guide plate.

5. Release the pilot operated check valve from the bottom of the cylinder.

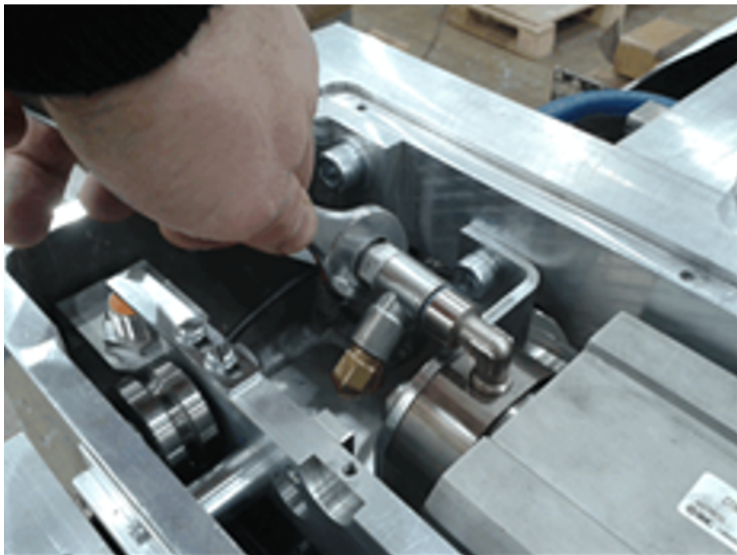


Figure 9.11-14 Example of an technician removing the pilot valve.

6. Loosen and remove the four hex-head bolts and spacers that connect the carriage plate to the pillars. Be sure not to lose the spacers.



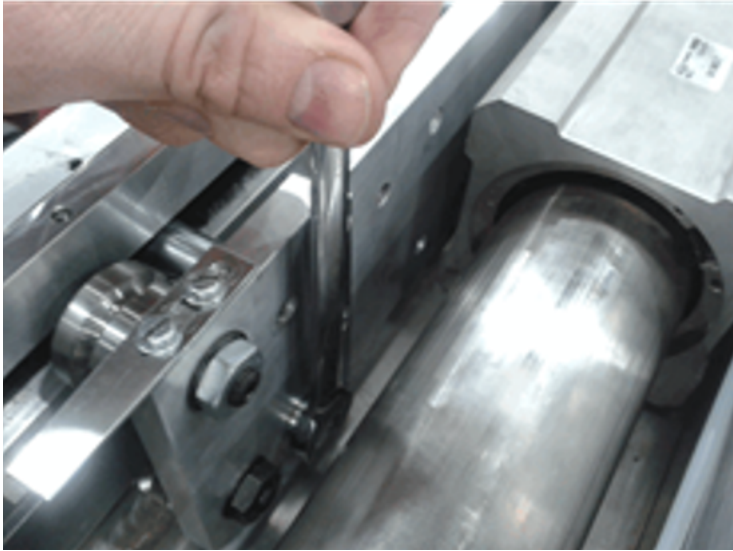


Figure 9.11-15 Examples of an technician removing the hex head bolts.

7. Push the cylinder body out of the way of the carriage plate.



Figure 9.11-16 Example of a technician pushing the cylinder body away to the opposite end.

8. Release the locking nuts





Figure 9.11-17 Example of a technician releasing the remaining fasteners.

9. Rotate the upper eccentric rollers to allow the carriage plate to be removed



Figure 9.11-18 Example of a technician rotating the eccentric rollers.

10. Manoeuvre the carriage plate off the Vee-rail by tilting as shown and then remove.





Figure 9.11-19 Example of a technician releasing the carriage plate.

The Vee rail is now accessible and able to be removed, by removing the holding fasteners.



Figure 9.11-20 Example of the Vee rail area cleared for access.

Follow the instructions in reverse to re-install the carriage plate.

To remove the cylinder from the assembly, release the air connections and remove the four cap-head screws from either end. The cylinder can then be removed.

## Vee-Rail and Cylinder Installation Instructions

This procedure is written on the understanding that anybody following it, is competent to do so. Using Drawing 417G155

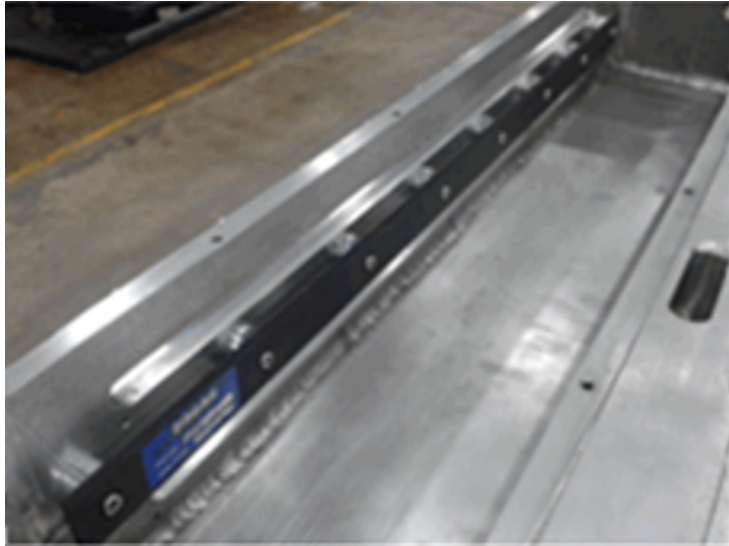


Figure 9.11-21 A mounted double edge Vee slide rail.

1. Mount the double edge Vee slide rail (B18) as shown, the spacers sit on a step on the inside face of the aluminium box, using spacers (M47) and screws (S15)  
The open/close sensors can be fitted at any time of the build process, but it is easier at this stage of the assembly. Use securing cable clips are used to secure the electrical wiring.

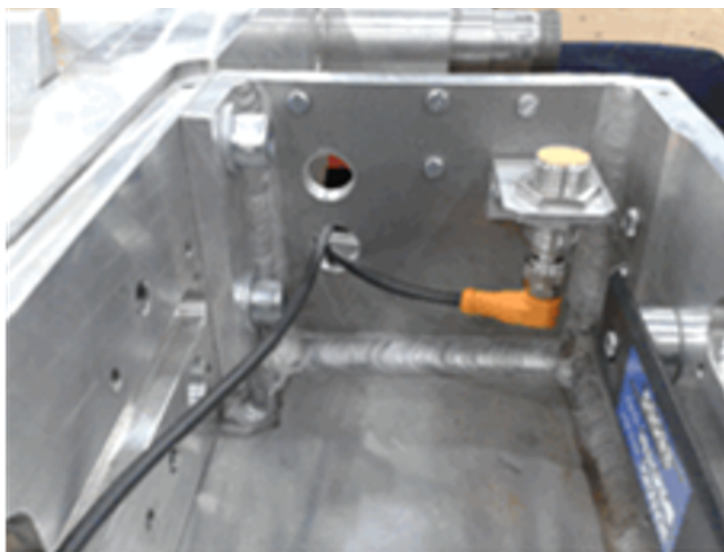
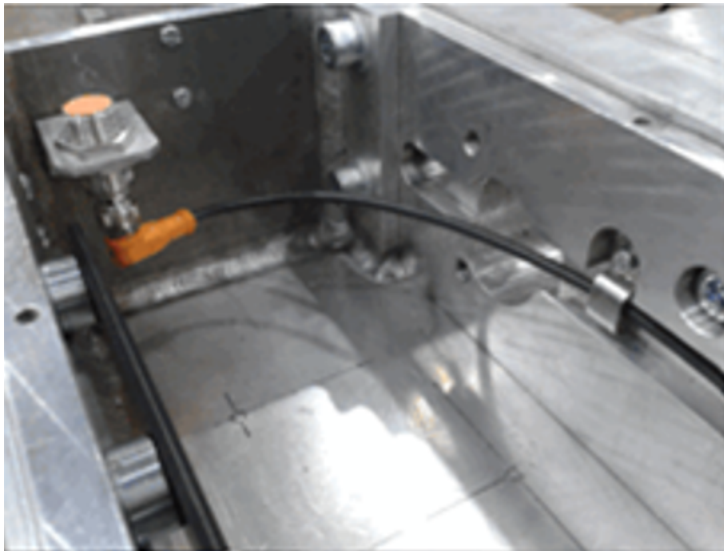


Figure 9.11-22 Position sensors located inside the enclosure

Install the concentric (B16) and eccentric rollers (B17) onto the carriage plates (M24 & M25). The concentric ones at the bottom of the plates.

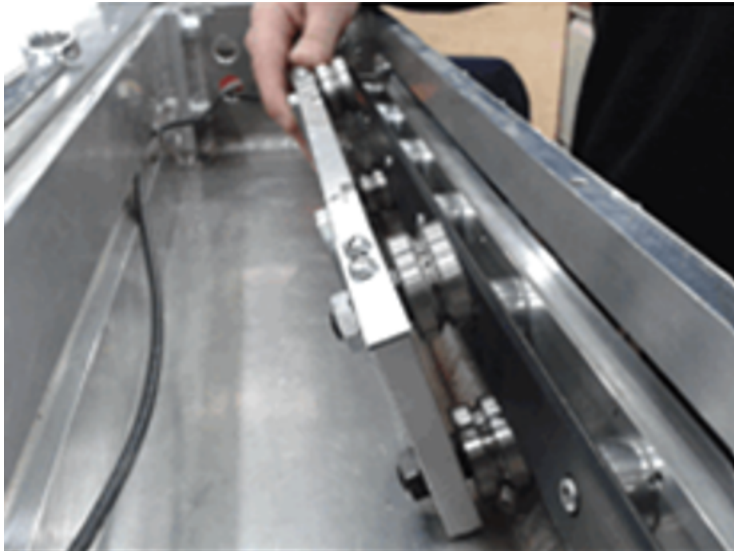


Figure 9.11-23 Carriage plate with rollers attached.

2. To secure the carriage plate (M24 & M25) to the rails, rotate the eccentric rollers to the highest position. This will give enough space to manoeuvre the rollers over the rail from the bottom.

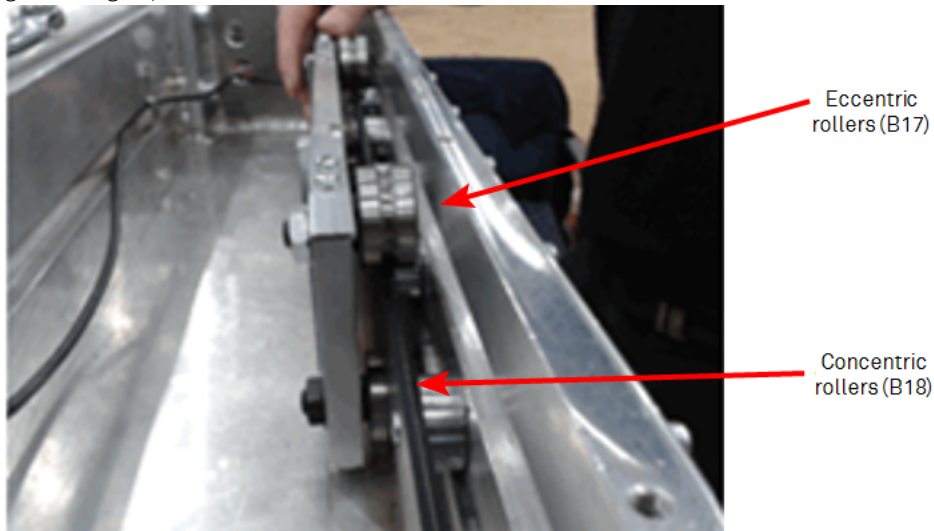


Figure 9.11-24 Example of a carriage mounted over the rail.

3. Locate the rollers to the rail by rotating the eccentric rollers (B17) using a 5mm Allen key until all rollers are in contact with the rail.
4. Fasten them in place with a spanner on the locking nut, to ensure that the eccentric roller doesn't rotate further. The tightening motion should be with the Allen key stationary in position.

Figure 9.11-25

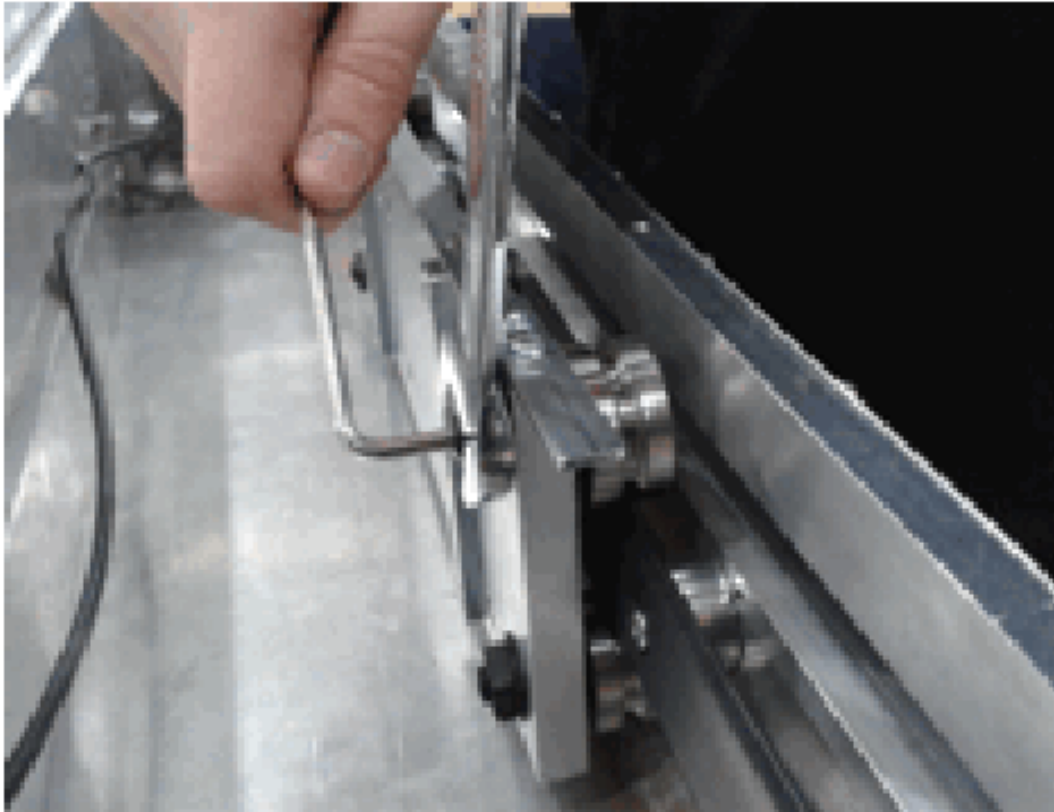


Figure 9.11-26 Example of a technician locking up the rail rollers.

5. Check that the carriage can slide up and down the rail.
6. Ensure that the rollers are not over-engaged that they restrict the carriage from sliding smoothly, yet not too loose that rattling can occur.
7. Ensure the brush strip guide brackets are fitted on the outside of the enclosure before continuing this procedure



Figure 9.11-27 Example of a fitted carriage plate.

8. On the inboard shutter side plate (M15) insert the guide pillars (M43) into the counter-bored holes in the shutter side plate (M15) with the slots facing upwards corresponding with the example. Locate the bolts that are used to secure them in place and grub screws to keep them at the correct angle/position.





Figure 9.11-28 Example of the shutter side plate (M15) with guide pillars attached (M43)

9. Insert the guide pillars (M43) through the openings in the bearing rail support and locate the pillars into the counter-bored holes on the carriage plate (M24/M25).

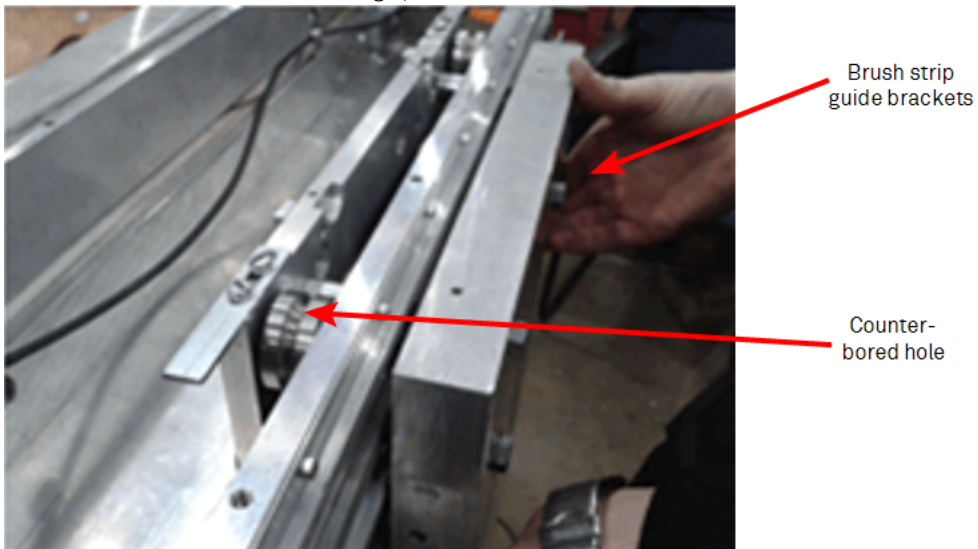


Figure 9.11-29 Example of a technician offering the side plate the assembly.

10. Locate the guide plate (M43) to the pillars using the 4 spacers and hex-head bolts.



Figure 9.11-30 Example of a technician locating the fasteners

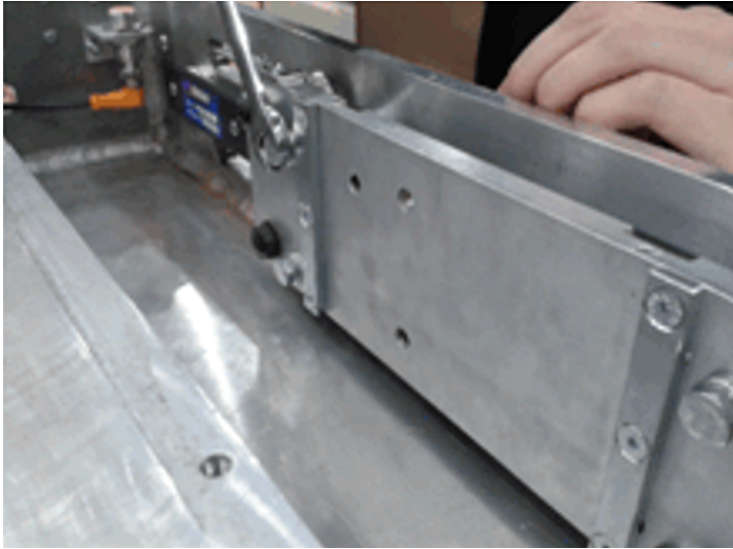


Figure 9.11-31 Example of a technician tightening the fasteners

11. Locate the cylinder in the shield enclosure and secure with cap-head screws and spring washers. Take care not to trap the sensor wires.



Figure 9.11-32 Example of a technician locating the cylinder in the shield enclosure.



Figure 9.11-33 The cylinder locating cap-head screws, also showing the routing of the sensor cable